

Manu Mathew Jiss

AI Software Engineer | Developer Experience

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Education

[University of the Pacific](#), Stockton, CA

May 2026

Master of Science in Computer Science

[APJ Abdul Kalam Technological University](#), Kerala, India

March 2023

Bachelor of Technology in Computer Science and Engineering

Technical Skills

Machine Learning & AI:	Python, PyTorch, TensorFlow, Scikit-learn, supervised and unsupervised learning, deep learning, model training, evaluation, optimization, feature engineering, NLP, computer vision, LLMs, Transformers, RoBERTa, CLIP
LLM & Retrieval Systems:	Generative AI, RAG, AI Agents, LangChain, LangGraph, Hugging Face Transformers, LoRA, PEFT, FAISS, Sentence Transformers, vector search, TextRank reranking, multilingual embeddings, OpenAI API, Anthropic API
Data & ML Pipelines:	Data collection, preprocessing, data labeling and validation, synthetic-data generation, Pandas, NumPy, PostgreSQL, pgvector, Supabase, MongoDB, Redis, ClickHouse, ETL pipelines, incremental ETL, data warehousing, metadata lineage
Deployment & MLOps:	FastAPI, Flask, Docker, AWS Lambda, Azure, Vercel, Railway, Git/GitHub, GitHub Actions, CI/CD, Linux, REST APIs, model serving
Engineering Tools:	SQL, JavaScript, TypeScript, C++, Bash/Shell, CUDA, OpenCV, ONNX Runtime, NVIDIA Jetson, ROS 2, Streamlit, Matplotlib, Plotly

Experience

- **Machine Learning Engineering Intern** May 2023 – July 2024
[Equalizer](#), Dubai, UAE (Remote)
 - Developed and deployed Python-based telemetry analytics pipelines using FastAPI, PostgreSQL, and Docker, applying anomaly detection and time-series forecasting to monitor distributed infrastructure and improve operational visibility.

Startup Experience

- **Co-Founder & Software Engineer** March 2026 – Present
[WhatToBuy](#) (AI Shopping Concierge)
 - Co-founded and launched a production AI shopping concierge using Next.js, TypeScript, FastAPI, Supabase, and Claude, engineering a multi-source commerce pipeline across Google Shopping, Amazon, and eBay that transforms natural-language scenarios into personalized, purchase-ready carts.
- **Co-Founder & Software Engineer** May 2025 – Present
[Stock Crusher](#) (GenAI Trading Platform)
 - Built and deployed an LLM-powered financial-intelligence system combining financial, news, and social-data APIs with multi-model reasoning; designed explainable recommendation workflows and graceful handling of partial model failures.
- **Co-Founder & Software Engineer** October 2025 – Present
[Meal Muse](#) (AI Recipe Assistant)
 - Developed an AI-powered recipe recommendation system using OpenAI models, FastAPI, PostgreSQL, semantic search, RAG, and vector retrieval; delivered personalized conversational recommendations through production backend services.

Graduate Research Assistant – University of the Pacific

- *Dr. Pramod Gupta* *December 2025 – May 2026*
 - Fine tuned a RoBERTa based emotion classification model on the GoEmotions dataset to improve multi class emotion recognition for AI powered reflective journaling.
 - Built [AIMoodDiary](#), a hybrid AI journaling platform combining transformer based emotion classification, LLM powered inference, lexical calibration, and longitudinal mood analytics; **paper accepted at IEEE ICMLA 2026**.
- *Dr. Solomon Berhe* *May 2025 – May 2026*
 - Built NLP pipelines using TF-IDF and RoBERTa to analyze 542 Reddit discussions; compared Naive Bayes, Logistic Regression, and RoBERTa, developed author-community sentiment-divergence metrics, and created a [dashboard](#); **paper accepted at IEEE ICMLA 2026**.
- *Dr. Don Lee* *August 2025 - December 2025*
 - Developed SignSight, a CLIP-based traffic-sign recognition framework that achieved 38% higher accuracy than CNN baselines on manipulated signs, and deployed a F1TENTH autonomous-vehicle stack using Jetson Orin Nano, ROS 2, Docker, CUDA, and a ZED 2i camera; **paper under review at CVPR 2026 ([OpenReview](#))**.
- *Dr. Tapadhiri Das* *December 2024 – May 2025*
 - Built explainable machine learning pipelines using XGBoost and LIME for sentiment, toxicity, and account authenticity classification, achieving 96.9% accuracy.
 - Built a risk-profiling framework for bot, scam, spam, and real account detection; **paper accepted at IEEE SMC 2026 ([Paper](#))**.
- *Dr. Tapadhiri Das* *July 2025 – March 2026*
 - Developed LIMFADD, an LLM-assisted synthetic-data generation and annotation framework for Instagram account-authenticity research, producing labeled real, spam, scam, and bot examples.
 - Designed data collection and XAI-based validation pipeline; **paper accepted at IEEE IS-TAS 2025 ([Paper](#))**.

Selected Projects & Research

- B2B Sales Intelligence and Lead Prioritization Platform** *September 2026*
 - Building a B2B lead-prioritization pipeline that combines and enriches company data, then engineers firmographic, growth, technology, and buying-intent features for lead scoring with XGBoost.
- Endpoint-Sensitive Causal Audits of Mixture-of-Experts Safety** *August 2026*
 - Audited how routing interventions affect generated text and safety evaluation in OLMoE and Qwen3-MoE, separating next-token and attention-cache effects and comparing answer-prefix with complete-answer grading; found that text changes did not necessarily change complete-answer compliance labels; paper under review at ICLR 2027 ([OpenReview](#)).
- BookRAG — AI Book Learning & Retrieval Platform** *July 2026*
 - Built a reusable RAG pipeline for digital books using LangChain, token-aware chunking, multilingual embeddings, PostgreSQL, and pgvector. Preserved page-level metadata to support grounded retrieval and question answering.
- Intelligent Plant Monitoring Platform** *January 2026*
 - Developed an AI-enabled IoT platform using FastAPI, PostgreSQL, React, Docker, and Azure, building cloud-synchronized Raspberry Pi sensor pipelines, real-time analytics, and intelligent plant-monitoring decision support. ([GitHub Repo](#)).